Boomika S

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Technical Skills

Languages: Java, Python, JavaScript, C, SQL, HTML, CSS
Frameworks & Tools: FastAPI, Flask, React, Node.js, Celery, REST APIs, JWT, LangGraph, LangChain, Docker, Git, GitHub, GitHub Actions, Linux
AI/ML: LLMs, RAG, Ollama, Groq, OCR, Evals, Machine Learning, PyTorch, scikit-learn
Databases: PostgreSQL, MongoDB, MySQL, Redis, pgvector, ChromaDB
Spoken Languages: Tamil (Native), English (Professional)

Experience

DevOps & Infrastructure (Intern) Jun 2026 – Sep 2026
QiBR Technologies (Startup)

- Built an AI-powered interior design platform using React, FastAPI, PostgreSQL and Ollama, enabling AI-assisted room design, conversational interactions and design session management.
- Containerized 5+ services with Docker and shared the setup with a 10-member team, cutting environment setup from 3+ hours to about 10–15 minutes.

Projects

Multi-Agent Retail Decision Intelligence | *React, FastAPI, PostgreSQL, LangGraph* | GitHub

- Forecasts demand at P10/P50/P90 and flags stockout/overstock risk, then recommends fixes through MILP optimization (Pyomo + HiGHS).
- Two LangGraph agents answer natural-language analytics queries and send replenishment approvals to managers on WhatsApp.

Autonomous Software Testing Platform | *React, FastAPI, MongoDB, LangGraph* | GitHub

- Turns an SRS document into test plans and scripts through a 9-stage pipeline of 8 agents, then runs real Playwright UI and HTTPX API tests.
- Classifies each failure into 7 root-cause categories and retries with bounded self-healing; exports Markdown/JSON/PDF reports.

AI-Powered Student Skills Assistant | *FastAPI, PostgreSQL, Groq LLM, WhatsApp API* | GitHub

- Answers natural-language questions about verified course completions, with read-only SQL guardrails (table allowlist, timeouts, row limits).
- Validated on a 60-case evaluation suite with 100% accuracy; available on a web dashboard and WhatsApp.

Smart Invoice Processing & Business Analytics | *React, Flask, MongoDB, PaddleOCR-VL, Docker* | GitHub

- Extracts fields from handwritten invoices with OCR, achieving ∼90% field accuracy on 10+ test invoices, with upload, review and search.
- Secured with Google OAuth, JWT, RBAC and audit logs; includes analytics dashboards and automated PDF reports.

Education

Bannari Amman Institute of Technology Tamil Nadu, India
B.E. in Computer Science and Engineering, CGPA: 8.38 *2024 – Present*

Achievements

Finalist – Next SSM Kernel Hackathon 2026: Organized by CDAC
Finalist – Snapdragon Multiverse Hackathon 2026: Organized by Qualcomm, Bengaluru
Finalist – NeuroHack, Shaastra'26 (IIT Madras): Built an ML solution on the DREAMER dataset
Finalist – Entrepreneurship Hackathon, STPI × Techniche 2025: IIT Guwahati
2nd Runner-Up – LudoHack 2025: Developed a browser-based game using p5.js
LeetCode: Solved 370+ problems with a contest rating of 1555